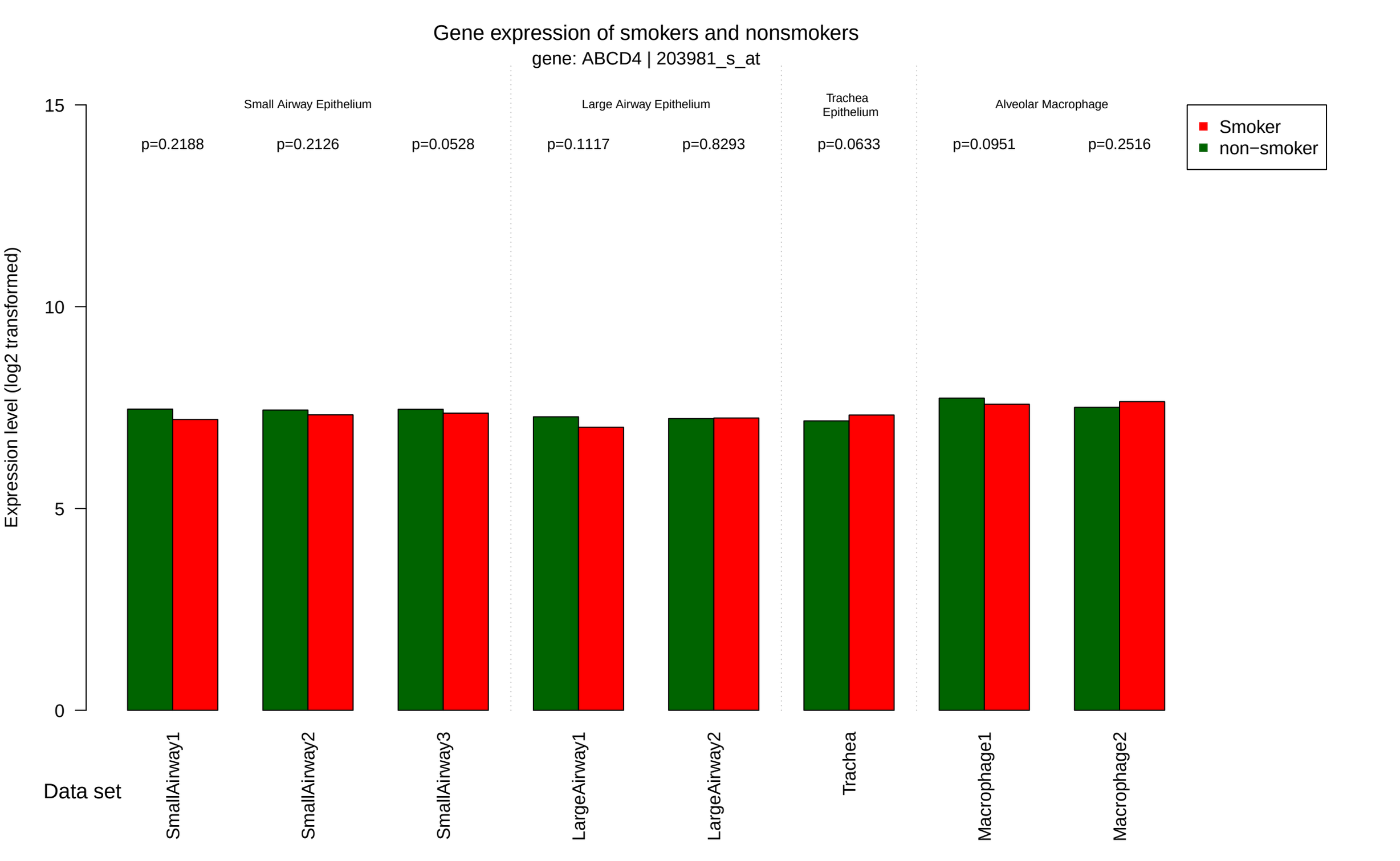
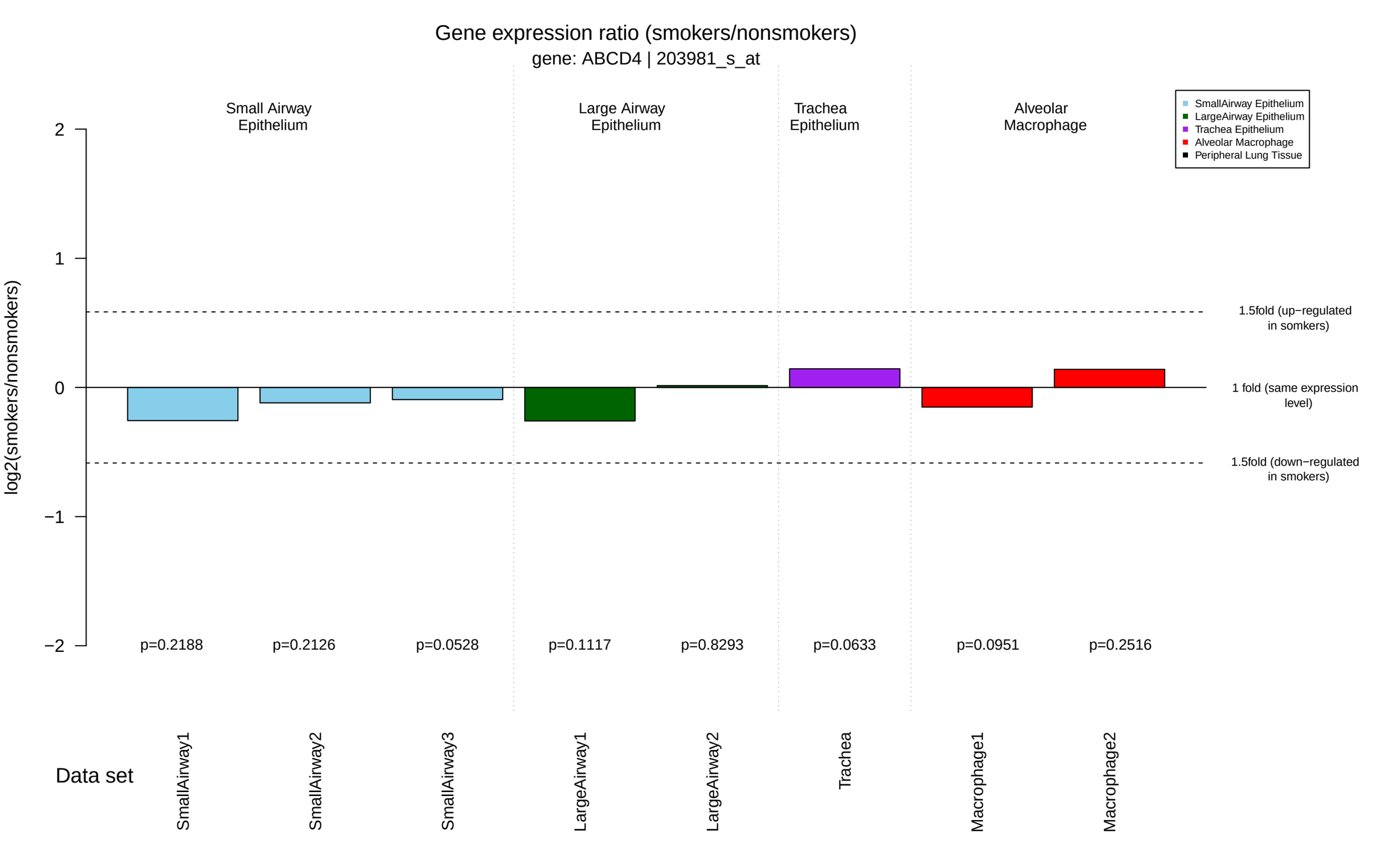
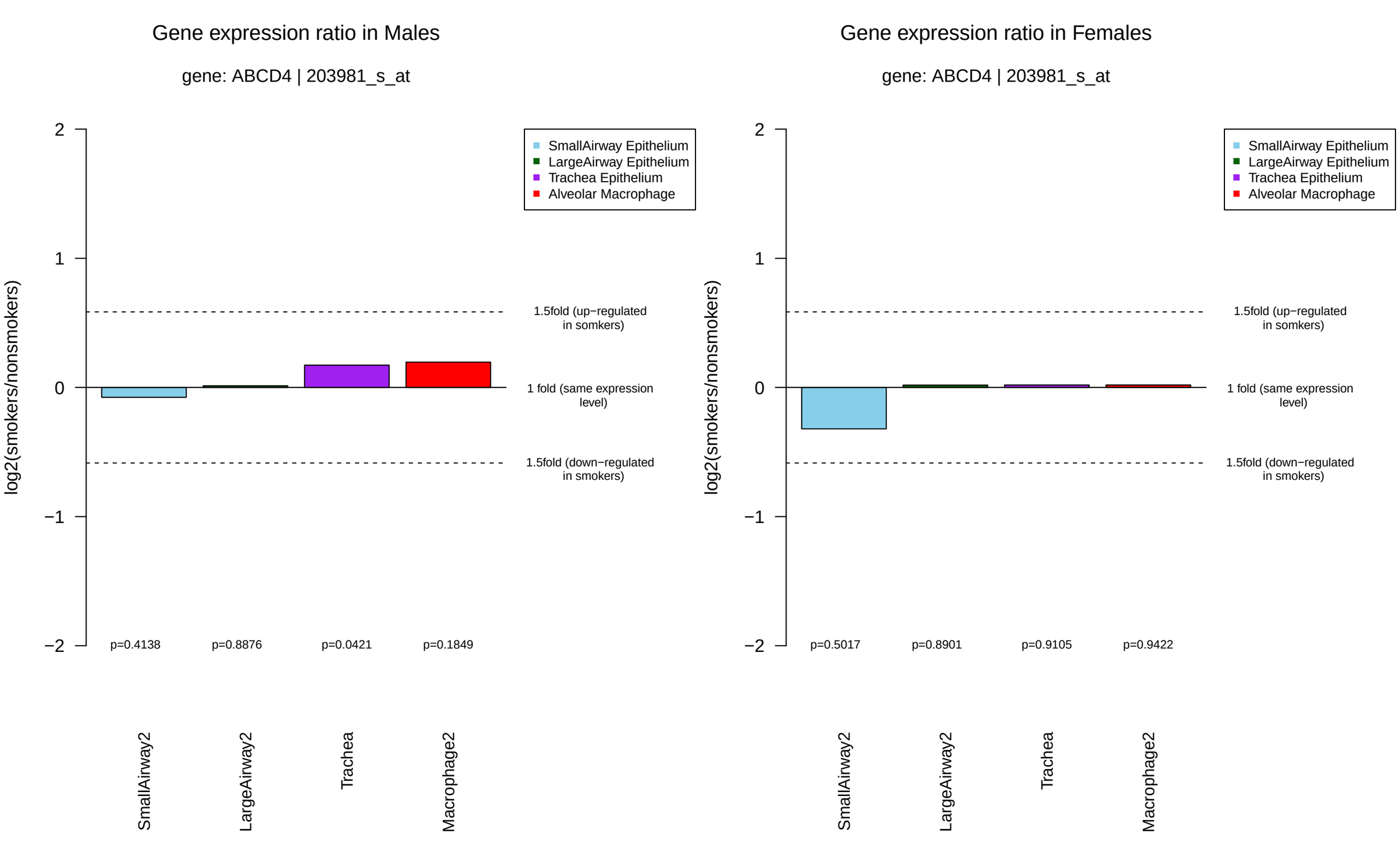
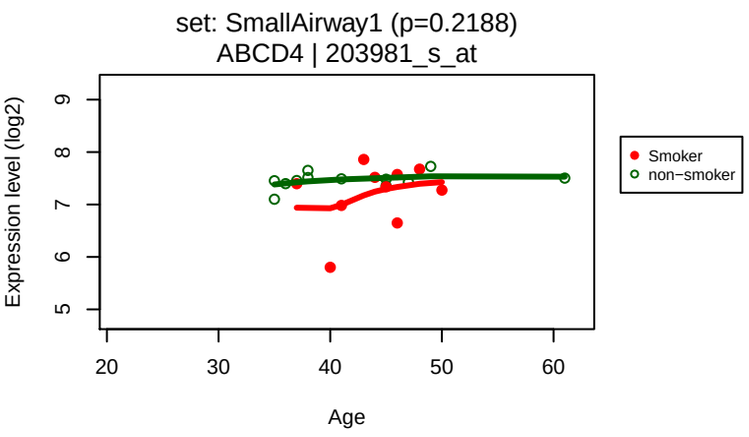
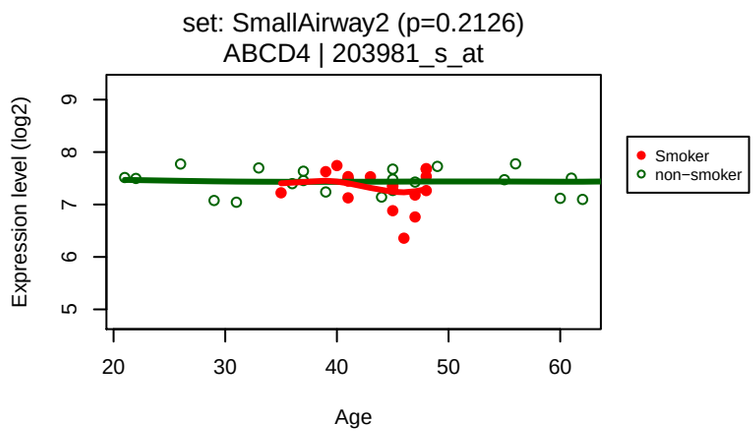
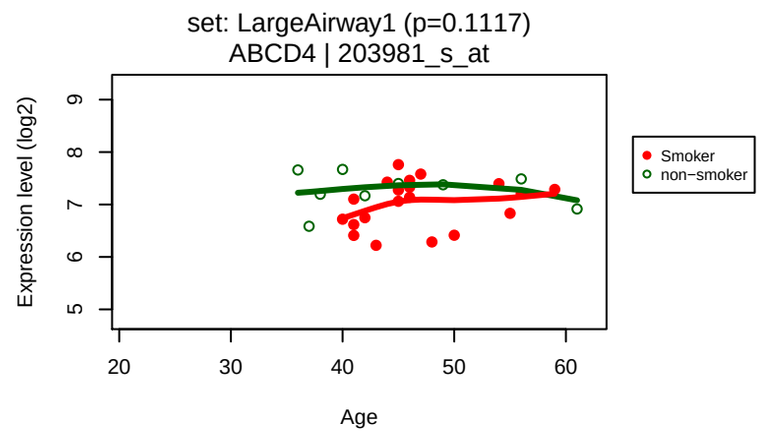

Gene expression of smokers and nonsmokers
gene: ABCD4 | 203981_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2188
p=0.2126
p=0.0528
p=0.1117
p=0.8293
p=0.0633
p=0.0951
p=0.2516
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

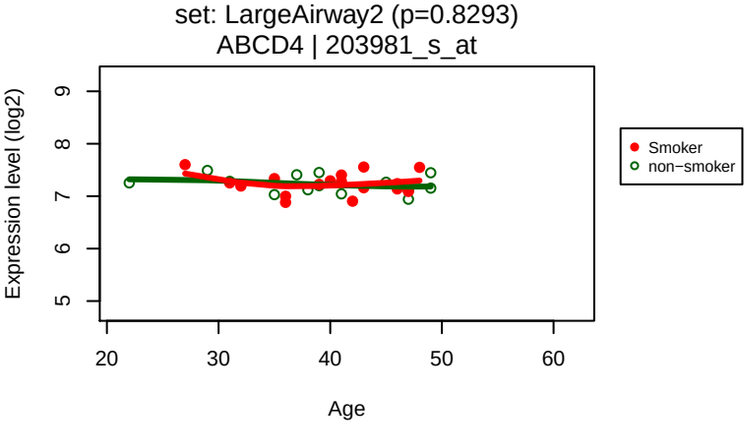

Gene expression ratio (smokers/nonsmokers)
gene: ABCD4 | 203981_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2188
p=0.2126
p=0.0528
p=0.1117
p=0.8293
p=0.0633
p=0.0951
p=0.2516
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

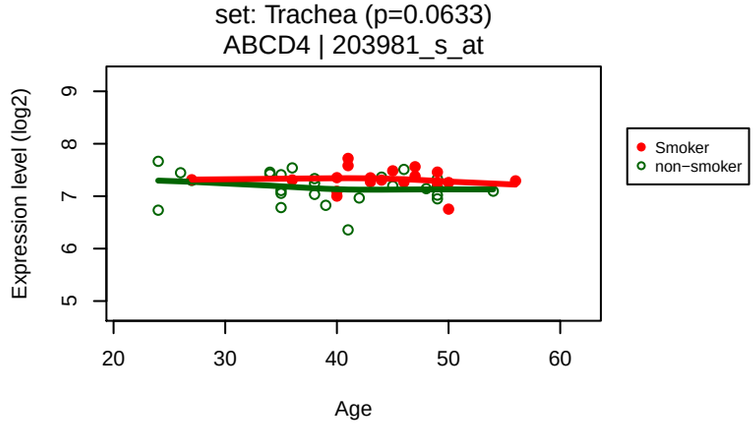

Gene expression ratio in Males
gene: ABCD4 | 203981_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4138
p=0.8876
p=0.0421
p=0.1849
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ABCD4 | 203981_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5017
p=0.8901
p=0.9105
p=0.9422
SmallAirway2
LargeAirway2
Trachea
Macrophage2

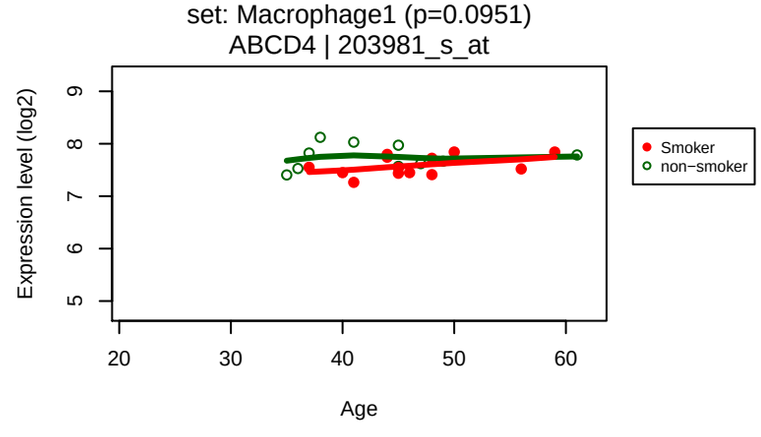

set: SmallAirway1 (p=0.2188)
ABCD4 | 203981_s_at
Expression level (log2)
Age
Smoker
non−smoker

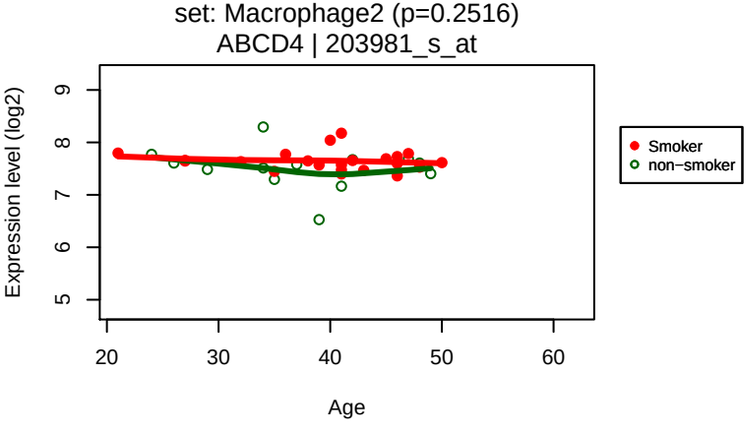

set: SmallAirway2 (p=0.2126)
ABCD4 | 203981_s_at
Expression level (log2)
Age
Smoker
non−smoker

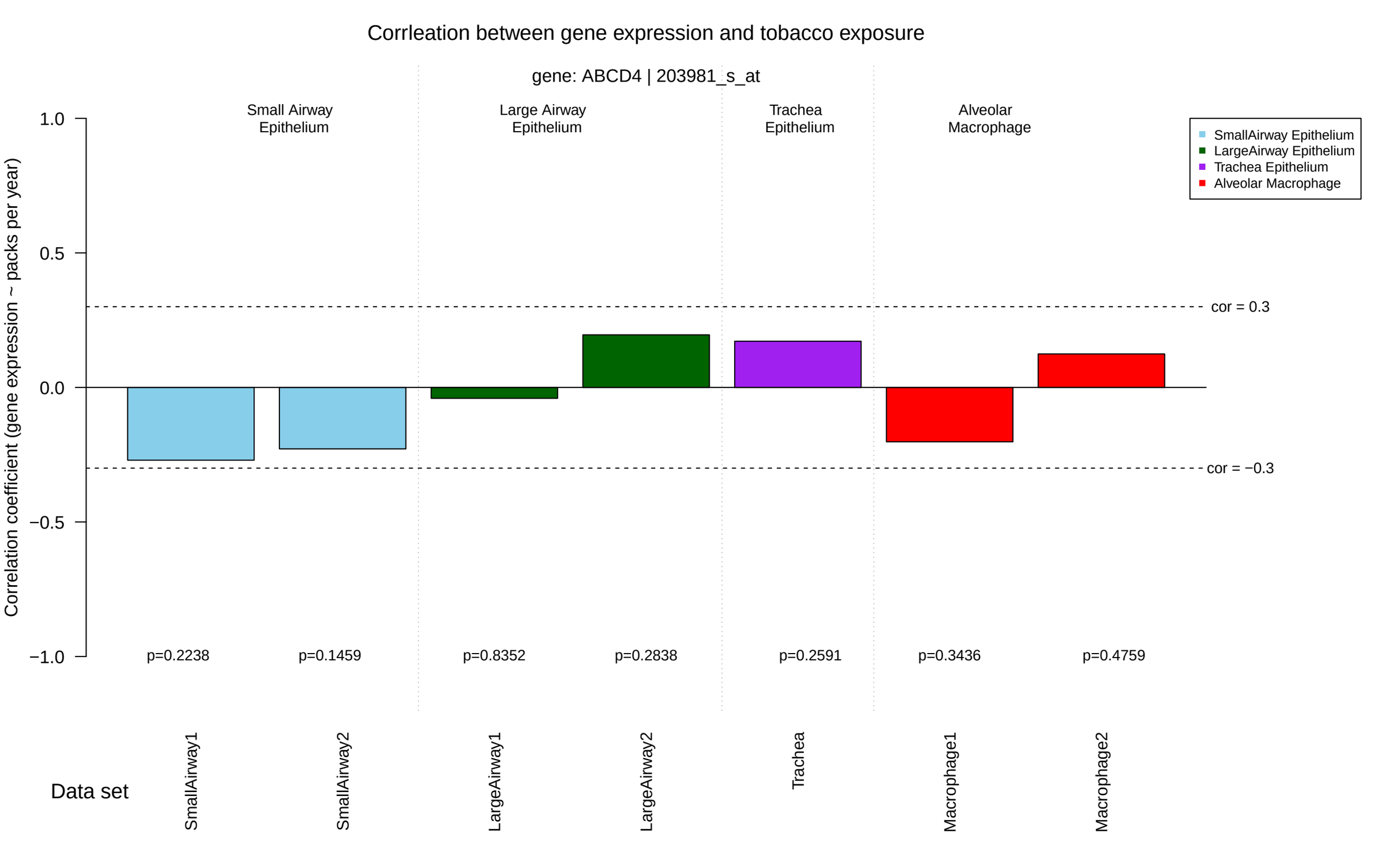

set: LargeAirway1 (p=0.1117)
ABCD4 | 203981_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8293)
ABCD4 | 203981_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0633)
ABCD4 | 203981_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0951)
ABCD4 | 203981_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2516)
ABCD4 | 203981_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ABCD4 | 203981_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2238
p=0.1459
p=0.8352
p=0.2838
p=0.2591
p=0.3436
p=0.4759
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2